

scope : L8–Unconscious v2

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About the Version 2 Revision

This paper is Version 2 of the previously published paper on L8. It reexamines the original design of L8 and reorganizes its definition and functional scope.

In the previously published version, L8 was primarily organized as “a cognitive layer that treats the cognitive structure as a whole as a single structure and references that structure in its entirety.” This description was not incorrect insofar as it represented one aspect of the functions associated with L8.

However, upon reexamination, it became clear that the previously published version did not sufficiently describe the design of L8, and that one aspect — “whole-structure reference” — had been expressed too strongly as the central definition of L8.

L8 had originally been designed to include the accumulation of materials available for cognition, such as knowledge, experience, and observation, as well as the organization of those materials into structures involving relationships among them. During the preparation of the previously published version, however, part of this design was not sufficiently reflected in the text, and the explanation was narrowed toward whole-structure reference. At the time of publication, the author also had not sufficiently recognized that this explanatory omission had occurred.

In particular, a part of the design related to why L8 had been referred to as “Unconscious” was not sufficiently reflected in the previously published version. This includes the aspect of L8 as an accumulative foundation that may contain materials that cannot be consciously and directly referenced. The previously published version did not position L8 as the unconscious in the psychological sense, but it did not sufficiently explain the relationship between the term “Unconscious” and the structural role assigned to L8.

Version 2 addresses this omission by reorganizing L8 as an accumulative foundation that handles materials available for cognition and the relational structures among them. These materials may serve not only as components for constructing new cognition, but also as a background against which current cognitive contents may be compared with existing accumulated materials. The relational structures among these materials are also not assumed to be fixed; this paper considers the possibility that they may be formed and reorganized through interaction with other cognitive layers.

Version 2 does not withdraw the description of “whole-structure reference” presented in the previously published version. Whole-structure reference is retained as one property of L8, while the overall positioning of L8 is reorganized and clarified to include the accumulation of

materials, relational structures, and interaction with other cognitive layers, which were not sufficiently described in the earlier version.

Accordingly, this revision is not positioned as a complete rejection of the earlier version or as a withdrawal of its individual claims. Rather, it is an update based on clarification of the definition and conceptual reorganization, intended to reexamine the original design and recover aspects that were not sufficiently reflected in the previously published version.

Chapter 0: Introduction

The Sephirotic names used in this paper are not intended in a religious or mystical sense. In SCOPE, they are borrowed as labels for identifying cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for inferring the cognitive structures of humans and AI. Here, a coordinate system does not mean a flowchart in which cognitive processing proceeds in a fixed sequence. Rather, it is a framework for organizing and inferring cognitive structure. The layers do not represent a hierarchy or temporal order.

In SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as “Malkuth.” What can be observed is the output itself, and it is not possible to directly know what cognitive structure operated internally. For this reason, SCOPE uses observable output as a clue to infer the cognitive structure behind it.

In human cognition, knowledge, experience, observations, and other material that are not currently being explicitly retrieved may appear to contribute as background to later judgment and understanding. Such accumulated material may serve as background not only when constructing a new idea, but also in situations in which one senses that “this may be possible” or “something about this arrangement seems inconsistent.” There are also cases in which something a person already knows is not referenced at the time, but is recognized as previously known after it is later presented. Such phenomena may not be fully captured by what is verbalized at that moment or by the explanations that are currently being explicitly referenced.

This paper addresses L8, one of these cognitive structures. L8 is the cognitive layer that addresses a foundation in which material available for cognition is accumulated as a structure of interrelations. This accumulation can serve not only as material for constructing new cognition, but also as a reference background when current cognitive content is compared with existing accumulation. However, this does not mean that L8 itself determines truth, falsity, or validity.

What L8 addresses is not only the existence of material. It also includes the structure in which each piece of material is positioned within its relations to other material. This structure is not fixed. Through new experience and interaction with other layers, the relationships and

positions among pieces of material may change. The definition and characteristics of L8 are discussed in greater detail from Chapter 1 onward.

L8 also has the property that the accumulated material and the overall structure of its interrelations may be referenced as background for cognition. However, this is not the central definition of L8. Rather, it is treated as one property associated with the fact that accumulated material and its relational structure exist as a background for cognition. The existence of the whole as background does not mean that all accumulated material is referenced simultaneously on every occasion.

L8 is not defined as either the starting point or the final destination of cognitive processing. SCOPE does not assume that cognition begins from any particular layer; rather, it treats cognition as a structure in which the layers mutually influence one another. L8 also does not refer to a particular biological organ, a specific region of the brain, or an actual storage area or implementation-level module. It is positioned within SCOPE as a cognitive foundation of accumulation.

This paper examines the characteristics and basic properties of L8, its relationships with other layers, and its possible functional correspondence in humans and AI, centered on its role as this foundation of accumulation. Whole-structure reference is treated as one of its properties. Some points remain at the hypothesis stage and are left as topics for future consideration.

Chapter 1: What Is L8?

L8 is a cognitive layer that addresses an accumulative foundation in which materials available for cognition are retained within relational structures.

This accumulation can serve not only as material for forming new cognition, but also as a reference background when current cognitive content is checked against existing accumulation. However, this does not mean that L8 determines whether content is true, valid, or correct, or selects a correct answer. The specific nature of checking within L8 is addressed in Chapter 4.

The term “material” here does not refer only to information that is consciously recalled at a given moment. Learned knowledge, past experiences, observations, information that has been seen or heard, information whose source can no longer be recalled, and information that is not usually recalled may all become materials handled by L8, as long as they remain available for subsequent cognition. L8 is not a place only for forgotten, unconscious, or nonverbal information. Consciously learned knowledge may also become material.

The retention of a material and the current availability of that material for reference within cognition are not the same. Even when a material exists, it may not appear in current cognition if it is not sufficiently connected to the relational structure being referenced at that time. Therefore, what matters in L8 is not only whether a material exists, but also how that material is positioned within relational structures.

The relationships and positions among materials may also change through new experiences, interactions with other layers, and other influences. Materials that were previously treated separately may come to be referenced within a single relation, or existing relations may be reorganized. However, this does not mean that L8 itself actively searches for correct answers, evaluates materials, or selects conclusions. L8 is not defined as a function that generates theories or judgments from the materials it contains.

In SCOPE diagrams, L8 in humans is labeled “Unconscious.” However, L8 as defined in this paper does not refer to the psychological concept of the unconscious itself. The label is a simplified expression of the fact that L8 may include not only content that is currently conscious, but also knowledge and experiences that are not usually referenced, as well as information whose source or purpose cannot be explicitly identified. Therefore, “Unconscious” is not the central definition of L8.

Humans and LLMs do not share the same implementation or formation processes. However, a functional correspondence may be examined in the fact that both may retain large amounts of material available for subsequent cognition or generation and may use those materials together with relations among their elements. Such accumulation may serve not only as material for forming new cognition or generation, but also as a reference background during checking. In SCOPE, this type of function is treated as a candidate for functional correspondence with L8. This does not mean that an LLM’s training data itself is identical to L8.

Likewise, even when cognitive content formed in other layers becomes positioned as part of the materials or relational structures available in L8, this does not mean that cognitive content moves from other layers to L8 in a fixed sequence and is then stored there. SCOPE does not treat the layers as operating in a fixed order, but as cognitive structures capable of mutually influencing one another.

In summary, L8 is an accumulative foundation that handles materials available for cognition and their relational structures. The existence of a material, the current referencing of that material, and the way it is positioned within relational structures are not the same. This accumulation may serve both as material for cognitive formation and as a reference background during checking. However, determining truth or validity, searching for or selecting a correct answer, and generating theories or conclusions are not themselves defined as functions of L8. L8 is therefore positioned not merely as a memory store or another name for the unconscious, but as a cognitive layer with a distinct role within SCOPE.

Chapter 2: Basic Characteristics of L8

Chapter 1 organized L8 as an accumulative foundation that handles cognitively available materials and the relational structures among them. This chapter organizes the basic characteristics that follow from that definition.

Materials in L8 are not treated as simply existing side by side. Each material may be positioned within relationships with other materials. In this paper, the terms “parent folder” and “subfolder” are used as explanatory metaphors for this relational structure. A larger grouping may be compared to a parent folder, while materials or smaller groupings associated beneath it may be compared to subfolders. However, these expressions do not imply the existence of literal folders in the brain or computer-like directory structures. The theoretical focus remains the relational structure itself; “parent folder” and “subfolder” are only metaphors intended to aid understanding.

Because such relational structures exist, the presence of a material and its accessibility to current cognition do not necessarily coincide. A material may be retained yet fail to appear in current cognition if no relevant relational path connects it to the grouping currently being referenced.

The relationships and positions of materials within L8 are not fixed. New experiences or interactions with other layers may alter existing relationships or create new relationships between materials that were previously separate. In addition, even without the addition of new material, changes in the relationships or positions among already existing materials may alter how they are referenced or how they appear within cognition.

An important point is that when new material is added, existing material is not necessarily erased or overwritten. Existing material may remain while its meaning, position, or manner of reference changes through its relationship with newly added material. Even when content that was previously regarded as factual has been retained as material, later information may reposition it as something such as “this was previously understood in that way” or “this is a type of misunderstanding that can occur.”

Using the metaphor of parent folders and subfolders, even when new information leads to a revision in understanding, existing material is not necessarily erased. Within the same parent folder, the subfolder to which the material belongs may change, or its relationships with other materials may change. In other cases, the parent folder itself may be reorganized. Which grouping functions as a parent folder, and what criteria are used to relate materials within it, may also differ across individuals. Updating within L8 therefore does not mean only deleting old material and replacing it with new material.

The material accumulated in L8 is also not necessarily correct. Incorrect information, inaccurate knowledge, or misperceived content may become material within L8 if it is retained in a form that remains available for later cognition. If such material is referenced in subsequent cognition, the error may be carried forward as a premise and may influence the direction of explanation, interpretation, judgment, or other cognitive processes. At the same time, new information or changes in relational structure may alter the position of that material or the way it is handled in later cognition.

The correctness of individual materials and whether the relationships among those materials correspond to reality must be considered separately. Individual materials may themselves be incorrect, while in other cases the materials may each be correct but the relationships among them may themselves be mistaken. It is also possible for individual relationships to hold while the overall structure does not correspond to reality. Incorrect relationships, biased

groupings, or relational structures that are inconsistent with reality may nevertheless form a subjectively coherent whole and produce a sense of “I understand,” “it connects,” or “it makes sense.” However, subjective coherence does not mean that the content is correct or consistent with reality. The formation of a relational structure, the appearance of internal consistency, and subjective conviction must be treated separately from truth or validity.

The range and variety of cognitively available materials accumulated in L8 may influence the range of cognitive candidates that can be constructed in other layers. As the range of available materials expands, the range of possible explanations, interpretations, judgments, abstractions, and other constructions based on those materials may also expand. However, a broad range or variety of materials does not mean that all of them will actually be referenced in cognition. Nor does a broader range or greater variety guarantee correct cognition, appropriate judgment, the ability to use those materials, or strong correspondence with reality. If incorrect materials or biased relational structures are referenced, cognition that does not correspond to reality may still be constructed from a broad range of materials. The range and variety of available materials must therefore be distinguished from how those materials are actually referenced and used.

L8 is not defined as a function that automatically determines whether materials are correct or whether their relationships correspond to reality and then corrects them. Changes in structure or the formation of subjective understanding are likewise not treated as the result of L8 searching for the correct answer, evaluating materials, or selecting conclusions.

The materials accumulated in L8 and their relational structures may also form, as a whole, part of the background of cognition, and this overall structure may be referenced in some cases. However, this does not mean that every accumulated material is simultaneously referenced on every occasion. Whole-structure reference is not the central definition of L8, but one characteristic that may arise from the relational structure within L8.

In summary, the basic characteristics of L8 are that materials are positioned within relational structures; that accessibility may vary depending on those relationships; that relational structures may change even without the addition of new materials; and that when new materials are added, existing materials are not necessarily erased but may instead change in position or role. The range and variety of available materials may influence the range of candidates that can be constructed in other layers, but they do not guarantee correctness or ability. To understand L8, it is therefore necessary to distinguish not only what materials exist, but also how those materials are positioned within relational structures and how they may become available for reference.

Chapter 3: L8 Changes

Chapter 2 organized how materials accumulated in L8 are positioned within relational structures, and how their relationships and positions may change. This chapter clarifies what is treated as “change” within L8.

L8 is not treated as a foundation that permanently retains fixed materials and fixed relational structures. New experiences, information, interactions with other layers, and other factors may add materials available for cognition. In addition, the relationships, groupings, and positions among materials that already exist may also change.

Change within L8 does not mean only the addition of new materials. Existing materials may remain while the grouping in which they are positioned, or the materials with which they are related, changes. Through such changes in relational structure, the way the same material is referenced in subsequent cognition may also change.

At the same time, the existence of a material in L8 and the ability to consciously and directly reference that material at a particular moment are not the same. Even if a material cannot be directly recalled at that moment, this alone is not treated as evidence that it has disappeared from L8.

The referenceability of a material may vary depending on the relational structure in which that material is positioned and on which relational structure is being referenced at that moment. A material that cannot be directly referenced from one relational structure may become referenceable when another related material or piece of information serves as a cue. Referenceability in this sense does not mean that L8 has a function that actively searches for or selects materials. In addition, a change in referenceability does not necessarily mean that L8 itself has undergone structural change.

In this paper, structural change within L8 includes the addition of materials and changes in the relationships, groupings, and positions of existing materials. A change in referenceability may arise because the relational structure itself has changed, or because a different relational structure is being referenced at that moment. Therefore, structural change within L8 and change in referenceability at a particular moment are treated as distinct.

These changes are not defined as resulting from L8 itself evaluating content, judging whether it is correct or incorrect, or actively modifying it. How interaction between L8 and other layers may contribute to such changes is addressed in the next chapter.

Chapter 4: L8's Relationship with Other Cognitive Layers

L8 is not treated as an accumulative foundation that operates independently and is complete in itself. Materials and relational structures accumulated in L8 may be used by other cognitive layers in forms corresponding to their respective cognitive functions. Conversely, cognitive content formed in other layers may later become positioned in L8 as part of the materials or relational structures available for subsequent cognition.

However, this does not mean that information passes through the layers in a fixed sequence and is ultimately stored in L8. In SCOPE, the cognitive layers are treated as capable of mutually influencing one another, and the relationship between L8 and the other layers is not

understood as a one-directional flow of input or output. Such interaction is not limited to one direction and may occur repeatedly.

A single material also does not necessarily correspond to only one cognitive layer. The same material may be involved in different cognitive functions across multiple cognitive layers. Therefore, particular information or experiences are not classified in advance as belonging exclusively to a single layer.

Materials accumulated in L8 may sometimes be used as they are in other cognitive layers, while in other cases multiple materials may be combined, transformed, or reorganized to form new cognitive content. When such newly formed cognitive content remains available for subsequent cognition, it may be positioned within an existing parent folder or subfolder, or added to the relational structure as a new grouping.

Changes in relational structure are not limited to changes in the positions of individual materials. An existing grouping itself may be repositioned as part of a broader new grouping, and relationships among groupings may thereby be reorganized. In this way, not only the materials in L8 but also the relational structures among those materials may change.

In this sense, the existence of a material in L8, its being referenced at a given moment, and its actual use in subsequent cognition are not the same. A material may exist in L8 without being referenced at that moment. Even when it is referenced, it does not necessarily have to be used in its existing form, and may instead be processed or reorganized through interaction with other cognitive layers.

In this paper, one function of L8 is treated as involving a check of whether newly formed cognitive content can be positioned within the relational structure currently being referenced. The checking described here does not determine whether that cognitive content is correct, corresponds to reality, or is valid. What is addressed is whether the cognitive content can hold a structural position within the relational structure currently being referenced.

Using the parent-folder and subfolder metaphor introduced in Chapter 2, this can be understood as the question of whether newly formed cognitive content can be positioned somewhere within the relational structure under the parent folder currently being referenced. Content that cannot be positioned under one parent folder may nevertheless become positionable when another parent folder, an intermediate grouping, or a broader grouping is referenced.

Accordingly, positionability is not determined by the cognitive content alone. It may also vary depending on which relational structure is being referenced at that time and through which intermediate relations the content can be positioned. Even the same material or cognitive content may be unpositionable within one structure but positionable within another when the range or grouping of the relational structure being referenced differs.

In addition, even when multiple materials are connected internally through intermediate relations, if those intermediate relations do not appear in output, the transition may appear from the outside to be a shift to an unrelated topic. This indicates that the relational structure

established internally and the range of relations that appear in observable output do not necessarily coincide.

Positionability, as used here, does not mean correspondence with reality or truth. Even when L8 contains incorrect materials or biased relational structures, cognitive content may still be positionable if a position for it can be established within those internal relational structures. Conversely, when no position can be established within the relational structure currently being referenced, the content may remain unpositionable within that structure. This checking is distinguished from determining real-world validity or truth, searching for materials, or selecting a correct answer.

The relationship between this checking of positionability, cognitive integration, and the emergence of Da'at is addressed in a separate paper.

Accordingly, the relationship between L8 and the other cognitive layers should not be understood as a fixed sequence or one-directional process, but as a relationship in which mutual influence may occur. L8 is an accumulative foundation of materials and relational structures that may be used by other cognitive layers. In addition, within L8, checking may occur as to whether newly formed cognitive content can be positioned in relation to the relational structure currently being referenced. However, this checking is distinguished from judgments of truth or validity, as well as from searching for materials or selecting a correct answer.

Chapter 5: The Relationship Between L8 and Malkuth

L8 is an accumulative foundation that handles materials available for cognition and the relational structures among them. However, it is not possible to directly observe what materials exist within L8 or how those materials are related to one another.

What can be observed in SCOPE is the output that appears in Malkuth as a result of cognitive processing. In the case of L8 as well, observable outputs such as speech, writing, and behavior are used as clues to infer what kinds of materials or relational structures may have been involved.

What can be inferred from output appearing in Malkuth is limited to hypotheses about part of L8. Even when detailed knowledge about a particular subject appears in output, that output alone does not establish that corresponding materials exist in L8. What can be inferred is limited to the possibility that some materials related to that subject were involved in cognition, or to hypotheses about part of the relational structure. In addition, the form in which something appears in output is not necessarily identical to the form of the material within L8. Output alone cannot be used to enumerate or reconstruct everything accumulated in L8, the extent of that accumulation, or the relational structures formed among those materials.

The fact that a material does not appear in Malkuth does not necessarily mean that it is absent from L8. L8 may include materials that are not consciously and directly referenced at that moment, as well as materials whose existence the cognitive subject cannot explicitly recall. Through their relation to other materials or through particular cues, such materials may later become directly referable. It is therefore necessary to distinguish between the existence of a material, its being referenced at a particular moment, and its actual use in cognition.

Furthermore, the entire contents of L8 are not necessarily fully accessible at all times, not only to an observer but also to the cognitive subject. This limitation in observability is related to part of the reason L8 has been referred to as “Unconscious.” However, L8 as defined in this paper does not refer to the unconscious in the psychological sense. Rather, it is positioned as a broader accumulative foundation that also includes materials that can be consciously referenced.

SCOPE therefore assumes this limitation when inferring part of L8 from results that appear in Malkuth. Neither L8 itself nor its entire contents can be reconstructed from Malkuth.

Chapter 6: Functional Correspondence of L8 in LLMs

Chapter 5 established that L8 itself cannot be directly observed and is instead inferred from outputs that appear at Malkuth. This chapter examines the possibility of functional correspondence between human L8 and LLMs.

Functional correspondence in SCOPE does not imply identity of internal structure or implementation. For each layer, the function defined in this paper is used as the unit of comparison in examining whether a functional correspondence may exist between humans and LLMs. Accordingly, combining multiple distinct functions after the fact into a single grouping and treating that grouping as the same function is not included in what this paper means by functional correspondence.

Humans and LLMs do not share identical internal structures or processes of formation. This paper therefore does not assume that a structure identical to human L8 exists within an LLM. However, in LLMs as well, there may be cases in which numerous pieces of information and structures available for subsequent generation are retained, and relationships among them may contribute to output generation. In this respect, a functional correspondence with L8 can be considered.

It is necessary here to distinguish among the training data itself, the internal structures formed through training, and the information actually available at a particular moment of generation. Information contained in training data is not necessarily stored internally as discrete information in the same form and directly retrieved during generation. Likewise, not all internal structures formed through training are necessarily used in the same way at every moment of generation.

Accordingly, what this paper treats as a possible functional correspondence with L8 is not the training data itself, but information and structures available for generation at a given time, together with the relationships among them. However, not all information available during generation is treated as corresponding to L8. This paper treats such information and structures as candidates for functional correspondence with L8 when they can function, within relational structures, as material for generation or as a referential background.

Information available for LLM generation is not limited to what was formed through training of the base model. Conversation history, information provided in the current interaction, externally supplied information, and stored information accessible to the system may also become reference targets during generation. This does not mean that all such information is retained through the same mechanism. What this paper addresses is not identity at the level of implementation, but a functional correspondence in which material available for generation and the relationships among that material may function as generative material or as a referential background.

Changes in the range treated as functionally corresponding to L8 in an LLM must be distinguished from cases in which new information merely becomes temporarily available for reference. For example, even if up-to-date information obtained through search, or information available only within a particular interaction, is used in generation, this alone is not treated as an update to the range regarded as corresponding to L8. By contrast, when changes to the base model or to stored information that remains available for subsequent generations produce persistent changes in the available material or relational structures, the range treated as functionally corresponding to L8 may also be considered to have changed. However, not every model update or system update necessarily changes that range.

At the same time, it may not be possible from the outside to fully enumerate all information and structures that were available within an LLM at a particular moment of generation. An inability to enumerate them does not mean that they do not exist or were not used in generation. Conversely, the fact that they cannot be enumerated does not itself constitute evidence that they exist or were used. This paper therefore limits itself to inferring the possibility of functional correspondence using observable outputs and available information as clues.

In addition, the fact that certain information is available for generation is not the same as that information actually being referenced in a particular generation. Nor does information that was referenced necessarily appear directly in the output. This corresponds to the distinction made for human L8 between material being present, being referenced at a particular moment, and being actually used in subsequent cognition.

In summary, this paper treats, among the information and structures available for generation in LLMs, those that can function within relational structures as generative material or as a referential background as candidates for examining functional correspondence with human L8. This does not mean that humans and LLMs share the same cognitive structure, that LLM training data itself is L8, or that any particular technical component corresponds to L8. What SCOPE addresses is the possibility of functional correspondence between humans and LLMs despite their differing internal structures.

Chapter 7: Functional Boundary Between L8 and L7

L8 is an accumulative foundation that handles materials available for cognition and the relational structures among them. L7, by contrast, handles high-abstraction, high-compression cognition formed through the accumulation of experience. Because both can relate to previously acquired information and experience, they may appear to overlap at first glance. However, the objects they handle and the conditions under which they are formed are different.

The materials handled by L8 are not limited to those derived from experience. Knowledge learned consciously, past experiences, observations, information that has been seen or heard, and other material that remains available for later cognition may become material in L8. It is also not a necessary condition that explicit attention was directed toward the information at the time it was acquired. What is important in L8 is not the route through which the material was acquired, but whether it is positioned within a relational structure as material that can be used in later cognition.

By contrast, the cognition handled by L7 is experience-derived. Here, experience does not mean merely having been exposed to information or retaining something as knowledge. It includes processes formed through engagement with a target, repetition, observation, and similar forms of experience. Through the accumulation of such experience, L7 may form high-abstraction, high-compression cognition such as sensations, insights, and highly abstract concepts.

The term experience-derived does not mean that the resulting cognitive content must always be explicitly conscious. A person may not always be able to verbalize the sensations or insights formed in L7. The relevant distinction is not whether the resulting cognition can be consciously accessed, but whether its formation involved the accumulation of the subject's own experience.

This does not mean that a single experience has no influence at all on the formation of L7. Even one experience may contribute to later cognitive formation. However, the defining feature of L7 is not the retention of an individual experience itself, but the formation of high-abstraction, high-compression cognition through accumulated experience. Accordingly, this paper does not establish a fixed threshold based on the number of experiences, such as defining L7 as beginning after a particular number of repetitions.

Because of this difference, information related to the same event may be treated as serving different functions in L8 and L7. In L8, information or the experience itself may be positioned as material available for later cognition. In L7, what is handled may instead be sensations, insights, or highly abstract concepts formed through the accumulation of such experience. In other words, L8 concerns what is accumulated as material available for cognition, whereas

L7 concerns what kind of high-abstraction, high-compression cognition has been formed through accumulated experience.

However, this distinction does not imply that processing proceeds in a fixed sequence from L8 to L7. SCOPE does not treat its layers as a fixed processing order. It therefore does not assume a linear flow in which material is first accumulated in L8, then processed in L7, and then returned to L8. L8 and L7 are treated as distinct functions that may influence one another, as may the other cognitive layers.

In summary, L8 handles materials available for later cognition and their relational structures without limiting their origin to experience. L7, by contrast, handles high-abstraction, high-compression cognition formed through the accumulation of experience. This functional difference distinguishes the two.

Conclusion

This paper has reorganized L8 as an accumulative foundation that handles materials available for cognition and the relational structures among them.

In the previously published version, the property of L8 as “referencing the cognitive structure as a whole as a single structure” was described as its central feature. This paper does not withdraw that property. Rather, by reorganizing the underlying foundation of the materials handled by L8 and the relational structures among them, whole-structure reference is repositioned as one property of L8.

The materials that may be included in L8 are not limited to knowledge that can be consciously referenced. They may also include experience, observation, information encountered in the past, and contents formed through interaction with other cognitive layers, as well as other diverse materials that may later become available for cognition. These materials do not merely exist independently; they may also be related to one another, and those relational structures themselves may change.

For this reason, L8 is neither a simple memory store nor the unconscious in the psychological sense. Nor is it a mechanism that determines truth or falsehood, or an agent that integrates cognition as a whole. The materials and relational structures handled by L8 may be used, processed, and reconstructed through interaction with other cognitive layers. Contents formed through such interactions may later become materials available for cognition, but this does not imply a fixed processing route.

L8 itself cannot be directly observed. What can be observed in SCOPE is the output that appears in Malkuth, and what can be inferred from that output is limited to partial hypotheses about what kinds of materials or relational structures may have been involved. The entire contents of L8 cannot be reconstructed from output, and the fact that a material does not appear in output is not, by itself, sufficient grounds for concluding that the material does not exist.

This paper also does not assume that humans and LLMs possess identical internal structures or implementations. Instead, the function defined for L8 is used as a unit of comparison, and the possibility of functional correspondence is treated as a hypothesis with respect to the existence of materials and relational structures available for cognition or generation, and their possible use in subsequent processing. This correspondence does not mean that L8 is identical to training data itself, nor does it mean that humans and LLMs share the same L8.

On the basis of the above, this paper does not add a new function to L8. Rather, it is positioned as Version 2, which recovers design content that was not sufficiently expressed in the previously published version and reorganizes the central positioning of L8. While retaining the existing description of “whole-structure reference,” this paper places it back within the positioning of L8 as an accumulative foundation that handles materials and relational structures, thereby clarifying the role of L8.

The relationship between L8 and Da’at, as well as the details of how cognitive structures involving L8 become connected to output, are outside the scope of this paper.